

MinImageScorer: Error-Driven Difficulty Scoring for Guided Copy-Paste Augmentation in Agricultural Object Detection

[FILL: Author Name¹, Supervisor Name¹]

[FILL: Department of XXX, University of XXX, Country]

Abstract—We propose MinImageScorer (MIS), an error-driven framework that scores per-object and image difficulty by combining model confidence and localization penalties on training-set failure signals. Validated on MinneApple and Global Wheat Head (GWHD), MIS reliably ranks objects and images by difficulty and correctly identifies domain- and size-level failure patterns without requiring any additional information. When used to guide copy-paste augmentation, MIS-selected objects outperform random selection on MinneApple ($\Delta AP = +0.013$); however, both strategies fall below baseline. On GWHD, domain shift renders copy-paste ineffective regardless of selection strategy. These findings establish MIS as a reliable diagnostic for structural difficulty in agricultural detection, while also revealing the conditions under which copy-paste augmentation is unlikely to help.

Index Terms—object detection, copy-paste augmentation, difficulty scoring, agricultural detection, small objects, domain shift

I. INTRODUCTION

Agricultural object detection benchmarks pose structural challenges that are qualitatively different from general-purpose datasets such as MS COCO. MinneApple [1] is dominated by densely packed, sub-resolution apple instances where detection failure is driven primarily by scale rather than appearance. Global Wheat Head Detection (GWHD) [2] aggregates images from seven geographic domains, introducing severe cross-domain visual variation that causes systematic performance drops on minority-domain images. These two failure modes—small-object resolution limits and cross-domain distributional shift—are well-documented by the respective dataset authors, yet no prior augmentation method explicitly diagnoses or targets them all at once automatically.

Copy-paste augmentation has demonstrated small gains on MS COCO [3], [4], where objects span a broad size distribution and images are largely within-distribution. Applied blindly to structurally constrained datasets, however, copy-paste risks compounding the very failure modes it aims to address: pasting sub-resolution objects amplifies scale noise, and pasting objects across domains amplifies distributional mismatch. No prior work provides an empirical criterion for assessing whether copy-paste is appropriate for a given dataset *before* augmentation is applied.

We address this gap with two contributions. First, we introduce **MinImageScorer** (MIS), a per-object difficulty score derived directly from model failure signals. We validate that MIS correctly identify dataset structural difficulty without

access to any additional information. Second, we conduct a controlled study showing that score-guided copy-paste outperforms random selection on the MinneApple dataset, while both strategies degrade below baseline. On the domain-shifted GWHD dataset, copy-paste is ineffective regardless of selection strategy. Taken together, these results show that MIS correctly identifies where a model fails, and that the failure mode of a dataset itself—not the selection strategy—determines whether copy-paste can succeed.

II. METHODOLOGY

A. Scoring Formulation

Let $G(I)$ be the set of ground-truth objects in image I and $\mathcal{P}(g) = \{p : \text{IoU}(p, g) \geq \tau\}$ the set of predictions matched to ground-truth object g . Denoting $c_p = \text{conf}(p)$ and $u_p = \text{IoU}(p, g)$, the *Object Difficulty Score* is defined as:

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} [\alpha(1-c_p) + \beta(1-u_p)], & \mathcal{P}(g) \neq \emptyset, \\ \alpha + \beta, & \text{otherwise,} \end{cases} \quad (1)$$

where α and β weight the confidence and localization penalties, respectively. Missed objects (no prediction with $\text{IoU} \geq \tau$) receive the maximum score $\alpha + \beta$. The *Image Difficulty Score* combines mean object difficulty with the false-positive rate:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I). \quad (2)$$

Object scoring uses a low confidence threshold ($\tau = 0.01$) to capture near-miss predictions; FP rate is computed at the optimal operating threshold to reflect realistic deployment errors. Weights α, β, w_1, w_2 are calibrated by maximizing Pearson correlation between S_{img} and per-image miss rate and FP rate on the training set, requiring no domain-specific prior knowledge.

B. Score-Guided Copy-Paste Pipeline

A baseline YOLOv8s detector is first trained on the target dataset. Inference on the full training set yields per-prediction confidence and IoU values, from which S_{obj} and S_{img} are computed for every annotated object and image.

TABLE I
PEARSON CORRELATION OF S_{img} WITH PER-IMAGE MISS RATE AND FALSE POSITIVE RATE

Dataset	$r(\text{score, miss})$	$r(\text{score, FP})$
MinneApple	0.698	0.688
GWHD	0.816	0.818

Following the same pipeline as the author of [3] have done we will extend the dataset size by using Copy and Paste pipeline. The special of our implementation is that instead of choosing a number of random objects in a random image and copy-paste it multiple times in random locations with the same image, we will use the score to change the probability of selecting an object and a image to copy and paste.

Under our *score-guided copy-paste*, the probability to choose an object and a background image are sampled proportionally to their difficulty score via exponential weighting—objects with high S_{obj} are drawn more frequently, deliberately concentrating augmentation effort on instances the baseline model most consistently fails to detect or localize. This new data then used to retrain a fresh YOLOv8s model under identical hyperparameters, ensuring that any performance difference is attributable to selection strategy alone.

III. SCORE VALIDATION

A. Correlation with Detection Failures

We can see in Table I that all correlations are strongly positive, confirming that the score reliably captures image-level difficulty.

B. Domain and Size-Level Validity

Domain-level (GWHD). Table II show that the harder domains (lower AP) have higher Mean S_{img} . This indicate a strong relation between our scoring with GWHD failed mode.

Size-level (MinneApple). Table III shows that S_{obj} is higher for small objects than medium objects, which mean our scoring has correctly matched the known ap between these size categories.

Together, these results demonstrate that **MIS** can indentify the primary structural difficulty at both domain and object-size granularity—using only training-set model failure signals.

IV. AUGMENTATION EXPERIMENTS

A. Setup

We use YOLOv8s initialized from COCO pretrained weights. We will use copy and paste augmentation to increase the dataset size by 30%. For each new image, 2–8 new objects are chosen to be pasted, applied Scale jitter, random flipping following Ghiasi et al. [4]. The only difference between the two experimental conditions is whether source objects are sampled uniformly (*Random CP*) or proportionally to S_{obj} (*Score CP*).

TABLE II
CORRELATION OF MEAN S_{img} AND AVERAGE PRECISION ACROSS GWHD DOMAINS

Domain	AP	Mean S_{img}
rres_1	0.572	0.200
ethz_1	0.550	0.227
inrae_1	0.523	0.251
arvalis_3	0.466	0.294
usask_1	0.408	0.312
arvalis_1	0.403	0.331
arvalis_2	0.365	0.424

TABLE III
CORRELATION OF OBJECT DIFFICULTY SCORE S_{obj} WITH OBJECT SIZE ON MINNEAPPLE.

Dataset	Score Type	Small object (≤ 32 px)	Medium Object (≤ 96 px)
MinneApple	AP Score	0.300	0.592
	Object Difficulty Score	0.378	0.222

B. Results

MinneApple. Table IV show that Score-guided copy-paste outperforms random selection across sub-metrics, confirming that difficulty-weighted sampling directs augmentation toward higher-signal objects. However, both conditions fall below baseline indicates degraded localization precision from pasting objects without boundary blending. The AP_S drop we believe reflects that score-selected objects are predominantly sub-resolution and cannot be made more learnable through repetition alone—a constraint imposed by the dataset itself, not by the selection strategy.

GWHD. Score-guided and random conditions are statistically indistinguishable and both fall below baseline. Score-guided sampling concentrates objects from domain-minority sources, amplifying existing domain imbalance rather than correcting it. Feature-space analysis confirms that GWHD augmented images suffer great distributional displacement, consistent with the larger performance gap. This we believe indicated that a domain-shifted dataset, selection quality is irrelevant if the fundamental problem is covariate shift between augmented and test images.

V. CONCLUSION

MinImageScorer reliably quantifies per-object detection difficulty from model failure signals.

Score-guided copy-paste outperforms random selection on MinneApple ,confirming that difficulty-weighted sampling improves selection quality. Nevertheless, both strategies fall

TABLE IV
THREE-SEED MEAN COCO AP. **BOLD** = BEST COPY-PASTE CONDITION PER DATASET. ALL COPY-PASTE CONDITIONS FALL BELOW BASELINE, INDICATING DISTRIBUTIONAL SHIFT OUTWEIGHS SELECTION BENEFIT.

Dataset	Condition	AP	AP ₅₀	AP ₇₅	AP _S	AP _M
MinneApple	Baseline	0.439	0.771	0.452	0.300	0.592
	Random CP	0.415	0.741	0.427	0.279	0.563
	Score CP	0.428	0.759	0.440	0.286	0.581
GWHD	Baseline	0.507	0.922	0.491	0.136	0.501
	Random CP	0.486	0.905	0.456	0.118	0.480
	Score CP	0.484	0.905	0.455	0.098	0.477

below baseline on both datasets because the root failure modes—sub-resolution scale limits on MinneApple and cross-domain distributional shift on GWHD—are structural constraints that copy-paste cannot overcome when augmented images are displaced from the test distribution, as confirmed by backbone feature-space analysis.

The practical implication is precise: copy-paste is beneficial when objects are underrepresented but visually compatible with the test distribution; it is harmful when the failure mode is structural (resolution limits or domain shift). MIS provides a diagnostic criterion before augmentation: inspect the size and domain distribution of high-score objects. If they cluster in a regime the model cannot resolve regardless of training exposure, copy-paste will not help.

Future work should explore two directions. *Better blending system*—Better blending method that can blend the boundary of the pasted object with the background image to make the augmented image more natural and reduce the distributional shift. *Dynamic scoring*—recomputing object difficulty during training rather than fixing scores upfront—would allow augmentation to adapt as the model improves, avoiding continued emphasis on objects that become learnable mid-training. Both directions aim to keep augmented images within the test distribution, addressing the core failure mode this work identifies.

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